

OD Demand Compression as a Linear Encoder–Decoder

Folding, unfolding, and response-preserving zone aggregation

Companion to `compress_the_response.tex` (the intuition) and `superzone_design_principles.md` (the engineering). Here we make the aggregation a precise linear system and use it to *choose* the encoder.

1 The assignment response chain

OD demand $d \in \mathbb{R}^{n_{OD}}$, path flow $f \in \mathbb{R}^{n_P}$, link flow $v \in \mathbb{R}^{n_L}$:

$$d \xrightarrow{R} f = Rd \xrightarrow{B^\top} v = B^\top f = \underbrace{B^\top R}_H d, \quad v = Hd.$$

R is the OD→path split, B the path–link incidence, and $H = B^\top R \in \mathbb{R}^{n_L \times n_{OD}}$ is the **OD-to-link response**. Link cost $\tau = c(v)$; linearised about equilibrium, $\Delta\tau \approx C_v \Delta v$ with $C_v = \text{diag}(c'_a(v_a))$, so the OD-to-time response is $T = C_v H$.

2 Encoder (fold) and decoder (unfold)

Let $S \in \{0, 1\}^{N \times K}$ map each original zone to one of K super-zones ($S_{ik} = 1$ iff zone $i \in$ super-zone k). Because an OD is a pair, the OD folding matrix is the Kronecker square $Q = S \otimes S$, and with $d = \text{vec}(D)$,

$$\textbf{encoder:} \quad E = Q^\top = (S \otimes S)^\top, \quad \bar{d} = Q^\top d \quad (\text{super-zone OD}).$$

A decoder U unfolds the super-zone OD back to a loadable original OD, $\tilde{d} = U\bar{d} = UQ^\top d$, giving a linear autoencoder $P = UQ^\top$, $\tilde{d} = Pd$. Two decoders serve different goals:

reconstruction $U_R = Q(Q^\top Q)^{-1}$ (best $d \approx U_R Q^\top d$), **computational** U_C (load super OD (k, l) at representat

DTALite/TAPLite uses U_C . The *hierarchical* build (`superzone_hier.py`) realises U_C in the network itself: super-zones are the only centroids, every original zone is a through node, zero-cost connectors join super ↔ member, so demand keyed to super-zones loads through the members.

3 The right error: response distortion, not OD reconstruction

The quantity to preserve is the link flow, not the OD table:

$$v = Hd, \quad \tilde{v} = H\tilde{d} = HUQ^\top d, \quad \boxed{e_v = v - \tilde{v} = H(I - UQ^\top)d}.$$

“Lossless” means $Hd \approx HUQ^\top d$, i.e. $H(I - UQ^\top)d \approx 0$ — *not* $d \approx UQ^\top d$. Travel time follows, $e_\tau \approx C_v e_v = C_v H(I - UQ^\top)d$, bounded by $\|e_\tau\| \leq L_c \|e_v\|$ for locally Lipschitz c .

Corner-case theorem (verified). If $S = I_N$ (one super-zone per zone) then $Q = I$, the computational decoder U_C is the identity loading, $UQ^\top = I$, so $e_v = 0$ *exactly*. Empirically: Chicago Sketch and Chicago Regional at $S = I$ reproduce the full assignment to every decimal (link-volume $R^2 = 1.000$, 0% RMSE, identical VMT/TT/gap). This is the unit test for any encoder/decoder.

4 Low-rank structure and the corridor subspace

H is huge but low-rank: many OD pairs share corridors. With $H \approx U_s \Sigma_s V_s^\top$ and the corridor projector $P_s = U_s U_s^\top$, we need only the *dominant* modes preserved,

$$P_s H (I - U Q^\top) d \approx 0.$$

Empirically (Chicago Regional, $N = 1790$): the demand matrix D itself is rank ≈ 32 — $\text{SVD}(D)$ holds 50% of the energy in rank 1 (the gravity/size mode), 90% in 3, 99% in 32. So $K = 178$ super-zones give *far* more capacity than the response needs: **the encoder rank is not the binding constraint** — **the decoder U_C is** (intra-super demand it drops and the single representative it loads through).

5 Choosing the encoder S : experiment ($K=178$, 20 iters)

Five constructions of S , each run and scored by response distortion vs. full:

encoder S	link R^2	corridor (fwy) R^2	%RMSE	volume kept
geographic quantile grid	0.808	0.892	73	0.690
geographic k -means	0.802	0.951	76	0.656
demand-weighted k-means	0.849	0.966	66	0.708
low-rank OD-SVD embedding	0.838	0.952	68	0.703
flow-direction k -means	0.795	0.943	77	0.651

Recommendation: demand-weighted k -means (cluster zone centroids with weights = total zone demand). It wins every metric — corridor $R^2 = 0.97$ — because the rank-1 mode *is* the gravity/size pattern, and demand-weighting puts super-zone resolution exactly where the response is largest. The **low-rank OD-SVD embedding** (cluster origins by the top- r left singular vectors of D) is the principled runner-up and should overtake demand-weighting on networks where corridors are *not* aligned with demand density. Plain geographic grids/ k -means and direction signatures trail. All k -means variants beat the quantile grid on corridors (more compact clusters).

6 Design problem and the decoder lever

The full statement:

$$\min_{S, U} \|P_s H (I - U Q^\top) d\|_W^2 + \lambda I(S) + \mu C(S), \quad Q = S \otimes S, H = B^\top R, P_s = U_s U_s^\top$$

with $I(S)$ the intra-super-zone (dropped) demand and $C(S)$ a compactness/ interpretability penalty. Since the experiment shows the decoder dominates the residual at usable K , the highest-value next move is a **better U_C** : spread each super-zone’s loading across its member access points in proportion

to member demand (a soft, demand-weighted U_C) instead of a single representative — this directly shrinks the $I(S)$ term and the single-access concentration error, the two things the corner case removes.

Summary. Zone aggregation is a linear autoencoder on the OD object: encoder $Q^\top = (S \otimes S)^\top$ folds, decoder U unfolds, and the only error that matters is $e_v = H(I - UQ^\top)d$ — response, not reconstruction. The OD response is rank ~ 32 , so capacity is abundant; pick S by *demand-weighted* clustering (corridor $R^2 \approx 0.97$ at $K=178$), keep the low-rank OD-SVD embedding as the response-aware alternative, and invest remaining accuracy in a demand-spread decoder U_C .